# AI-Powered Land Pre-Purchase Verification System

**Main Problem Statement:** P5 — Citizen Pre-Purchase Verification

**Core Concept:** Help ordinary citizens check a property before buying it by analyzing available land records, ownership history, mutation information, registration data, and possible fraud indicators. The system gives a simple risk level and clearly explains why the property may be risky.

## 1. The Problem

A person buying land may not have enough legal or technical knowledge to understand every document. A seller may provide incomplete, conflicting, or misleading information. This can create risks such as ownership mismatch, double-selling, mutation conflicts, and legal disputes.

## 2. Proposed Solution

The citizen enters land information or uploads relevant documents. The system extracts and compares important information, checks for inconsistencies and suspicious patterns, and produces an easy-to-understand property verification report.

## 3. How It Works

| Step | System Action |
|---|---|
| 1 | Citizen enters District, Upazila, Mouza, Dag Number and Khatian Number, or uploads documents. |
| 2 | System collects available ownership, mutation, registration, legal-status and transfer information. |
| 3 | OCR and document extraction identify names, Dag, Khatian, area and other important fields. |
| 4 | Verification engine compares information across records. |
| 5 | Fraud detection checks for double-selling, ownership conflicts, suspicious transfers and inconsistencies. |
| 6 | Risk engine calculates a transparent risk level. |
| 7 | Citizen receives a simple report explaining the detected issues and recommended next verification steps. |

## 4. Main Features

• **Ownership Verification:** Compare seller identity with the available ownership record.

• **Double-Selling Detection:** Detect cases where the same land appears in multiple conflicting transactions.

• **Ownership History:** Show previous owners and transfer history as a timeline.

• **Mutation Verification:** Compare mutation information with ownership and registration records.

• **Legal Risk Check:** Flag available information about disputes, pending cases, acquisition or other land-status issues.

• **Document Upload:** Extract relevant information from deeds, khatian/porcha and mutation documents using OCR.

• **Risk Score:** Provide Low, Medium or High risk with the reasons behind the result.

• **AI Explanation:** Explain technical findings in simple Bangla or English.

• **Verification Report:** Generate a clear summary that the citizen can review before proceeding.

## 5. Example Scenario

**Seller:** Rahim
**Dag:** 125
**Khatian:** 456

The seller claims that Rahim is the current owner. The system finds that the available ownership record contains a different owner and that the same Dag Number appears in another transaction.

**Result: HIGH RISK — 91/100**

Detected issues:
1. Ownership mismatch
2. Possible double-selling
3. Conflicting transaction information

**Recommendation:** Do not proceed without further official verification.

## 6. AI and Fraud Detection Layer

The project combines rule-based verification with machine learning where appropriate. Clear conflicts such as duplicate ownership can be detected using deterministic rules. Anomaly detection can identify unusual transfer patterns. OCR and information extraction can convert documents into structured data. An AI assistant can explain the findings in simple language.

## 7. Suggested Technical Architecture

**Frontend:** React / Next.js
**Backend:** FastAPI
**Database:** PostgreSQL + PostGIS
**OCR:** PaddleOCR or Tesseract
**ML:** Scikit-learn anomaly detection and entity matching
**AI:** Local/open-source LLM for explanations and document reasoning
**Optional:** LangGraph for multi-step AI workflows

## 8. System Flow

**Citizen** → **Web App** → **FastAPI** → **Database / Documents** → **OCR & Extraction** → **Verification Engine** → **Fraud Detection** → **Risk Scoring** → **AI Explanation** → **Property Report**

## 9. Why This Approach Is Strong

P5 directly addresses the citizen's need to make a safer decision before purchasing land. P3-style fraud detection becomes the intelligence layer behind the system. This creates a practical product with a clear social impact while still providing strong AI/ML and backend engineering depth.

## 10. Important Prototype Limitation

For a hackathon prototype, use synthetic or authorized sample data unless official government data access is available. The system should not claim to provide official legal verification without authorized integration. The prototype should clearly present itself as a decision-support and risk-screening tool, not a replacement for official verification or legal advice.